Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01]

A functional equation constrains whole orbits, not isolated values. Compose transformations, preserve domains, and use rigidity when a polynomial or an infinite integer orbit cannot support the proposed behavior.

函数方程限制的是完整轨道，而非孤立函数值。应复合变换、保留定义域，并利用多项式或无穷整数轨道无法支持某些行为的刚性。

Compose input symmetries and classify value orbits

复合自变量对称并分类函数值轨道

Procedure

1. Compose the stated input transformations, carrying along any output sign or multiplier.

2. Identify genuine periods once the accumulated multiplier is one; integer periods combine by their gcd.

3. On integer residues, count the remaining reflection orbits, and check whether a sign relation forces a value to zero.

4. Construct a function satisfying every original relation to prove a proposed number of distinct values is attainable.

通用步骤

1. 复合题给自变量变换，同时保留输出的符号或倍数。

2. 累计输出倍数为一时得到真正周期；整数周期可取最大公因数。

3. 在整数余数上数剩余反射轨道，并检查带负号关系是否迫使函数值为零。

4. 构造满足全部原关系的函数，才能证明不同函数值数目可达到。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01] (continued / 续)

> Example. A real function obeys $f(-x) = -f(x)$ and $f(2-x) = f(x)$. What is the maximum number of distinct values at integer inputs?

> 例题：实函数满足 $f(-x) = -f(x), f(2-x) = f(x)$，在所有整数输入处最多有多少个不同函数值?

Worked solution

1. Composition gives $f(x+2) = -f(x)$ and hence period four. Oddness gives $f(0) = 0$, so every even integer has value zero.

2. Writing $f(1) = a$, odd integer values alternate $a, -a$. Thus at most the three values $0, a, -a$ occur.

3. The function $f(x) = \sin(\pi x/2)$ satisfies both original identities and realizes $0, 1, -1$, proving the maximum is three.

例题解析

1. 复合得 $f(x+2) = -f(x)$，因此周期为四。奇性给出 $f(0) = 0$，故所有偶整数处值为零。

2. 若 $f(1) = a$，奇整数处交替为 $a, -a$，所以至多有零、a、$-a$ 三值。

3. $f(x) = \sin(\pi x/2)$ 满足两原关系并实现零、一、负一，故最大值为三。

An upper bound on orbit classes still needs a function separating the permitted classes.

轨道分类只给上界，还需函数把允许分开的类别真正区分。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01] (continued / 续)

For a period P and reflection $r \mapsto c - r \pmod{P}$, fixed residues solve $2r \equiv c \pmod{P}$. If there are F fixed residues, the number of value orbits is $(P + F)/2$. A cosine centered at $c/2$ separates exactly the reflection pairs when no other relation is present.

周期为 P，剩余反射为 $r \mapsto c - r \pmod{P}$ 时，不动余数满足 $2r \equiv c \pmod{P}$。若有 F 个不动余数，轨道数为 $(P + F)/2$。无其他关系时，以 $c/2$ 为中心的余弦可恰好区分这些反射对。

Practice / 配套练习: AIME-2000-I-Q12 (WB p. 210)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 2 / 方法 2 [AL09-M02]

Extend a piecewise template by multiplicative scaling

按乘法缩放扩展分段模板

Procedure

1. Use $f(rx) = rf(x)$ to move any positive input into the defining interval $[1, r]$.

2. Write the exact piecewise formula on $[r^k, r^{k+1}]$ and check endpoint consistency.

3. For a target level, compare it with each scaled branch maximum before solving branches.

4. For a least input or a level-set length, combine the branch domains as well as the algebraic solutions.

通用步骤

1. 利用 $f(rx) = rf(x)$，将正输入缩放到定义模板区间 $[1, r]$。

2. 写出 $[r^k, r^{k+1}]$ 上的准确分段式，并检查端点一致。

3. 求给定水平时，先与各缩放分支最大值比较，再解方程。

4. 求最小输入或水平集长度时，必须将分支定义区间与代数解合并。

> Example. For $x > 0$, let $f(2x) = 2f(x)$ and $f(x) = \min(x - 1, 2 - x)$ on $[1, 2]$. Find the length of the set of $x \in [1, 8]$ with $f(x) \geq 1$.

> 例题：正数域函数满足 $f(2x) = 2f(x)$，在 $[1, 2]$ 上为 $\min(x - 1, 2 - x)$，求 $[1, 8]$ 内满足 $f(x) \geq 1$ 的集合长度。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 2 / 方法 2 [AL09-M02] (continued / 续)

Worked solution

1. On $[1,2]$, the maximum is $1/2$, so there are no qualifying points.

2. On $[2,4]$, $f(x)=\min(x-2,4-x)$ reaches one only at $x=3$, a set of length zero.

3. On $[4,8]$, $f(x)=\min(x-4,8-x)$. The condition is $5\le x\le 7$, so the total length is two.

例题解析

1. 在 $[1,2]$ 上，最大值为二分之一，没有合格点。

2. 在 $[2,4]$ 上，$f(x)=\min(x-2,4-x)$，仅在三处达到一，集合长度为零。

3. 在 $[4,8]$ 上，$f(x)=\min(x-4,8-x)$，条件为五至七，故总长度为二。

A smaller scale may contain no solution even when the branch equation has formal roots.

较小尺度可能根本达不到目标水平，即使形式方程有解。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 3 / 方法 3 [AL09-M03]

Expose finite-order composition and check all poles

找出有限阶复合并检查全部极点

Procedure

1. Compute a few exact compositions of a fractional-linear or affine map.

2. Prove a full composition returns the input, retaining every excluded denominator.

3. Reduce large iteration counts by this order only after proving the candidate orbit avoids the poles.

4. For affine complex maps $z \mapsto \omega z + c$, sum the geometric translation terms rather than tracking each point.

通用步骤

1. 对分式线性或仿射映射计算若干次精确复合。

2. 证明若干次复合回到输入，并保留每个禁用分母。

3. 只有证明候选轨道避开全部极点后，才能按阶约去大迭代次数。

4. 复仿射映射 $z \mapsto \omega z + c$ 应累加等比平移项，不必逐点追踪。

> Example. For $g(x) = 1/(1-x)$, determine all real inputs whose entire forward orbit exists, and the period of each such orbit.

> 例题：对 $g(x) = 1/(1-x)$，求所有完整正向轨道均有定义的实输入，并确定其周期。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 3 / 方法 3 [AL09-M03] (continued / 续)

Worked solution

1. The pole is one. Its only real preimage is zero, so zero and one must both be excluded.

2. For every other input, the map produces neither zero nor one. Direct composition gives $g^2(x) = (x-1)/x$ and $g^3(x) = x$.

3. A real fixed point would solve $x^2 - x + 1 = 0$, which has negative discriminant. Thus the allowed domain is $\mathbb{R} \setminus \{0, 1\}$ and every orbit has exact period three.

例题解析

1. 极点为一，其唯一实原像为零，因此零、一均须排除。

2. 其余输入的像既不为零也不为一。直接复合得 $g^2(x) = (x-1)/x, g^3(x) = x$。

3. 实不动点须满足 $x^2 - x + 1 = 0$，但判别式为负。因此允许域为 $\mathbb{R} \setminus \{0, 1\}$，每轨道恰为三周期。

A simplified composition can hide intermediate poles; domain checks belong to the unsimplified orbit.

化简复合式可能遮住中途极点，定义域必须按实际轨道检查。

For an affine complex map, $F^N(z) = \omega^N z + c\sum_{j=0}^{N-1}\omega^j$. If $\omega^N = 1$ and $\omega \neq 1$, the translation sum vanishes, proving finite order for the entire map.

复仿射映射满足 $F^N(z) = \omega^N z + c\sum_{j=0}^{N-1}\omega^j$。若 $\omega^N = 1, \omega \neq 1$，则平移和为零，整个映射具有有限阶。

Practice / 配套练习: AIME-2002-I-Q12 (WB p. 87), AIME-2008-II-Q09 (WB p. 70), AIME-2012-II-Q11 (WB p. 214)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 4 / 方法 4 [AL09-M04]

Use finite root orbits to force polynomial rigidity

用有限根轨道迫使多项式刚性

Procedure

1. Compare leading and constant coefficients of the polynomial identity.

2. Determine how every complex root generates further roots under the induced map.

3. Exclude any root modulus that would grow without bound in a finite root set.

4. Combine the product of root moduli and equality conditions to identify the remaining possible roots.

通用步骤

1. 先比较多项式恒等式的首项及常数项。

2. 确定每个复根在诱导映射下如何产生新根。

3. 根集有限，故排除会不断增大模长的根。

4. 结合根模乘积与等号条件，确定剩余可能根。

> Example. A nonempty finite set R of nonzero complex numbers is closed under squaring and satisfies $\prod_{z\in R}|z|=1$. Show every member is a root of unity.

> 例题：非零复数的非空有限集 R 在平方运算下封闭，且各元素模的乘积为一，证明每个元素都是单位根。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 4 / 方法 4 [AL09-M04] (continued / 续)

Worked solution

1. If some $|z| > 1$, repeated squaring gives unbounded moduli inside the finite set, impossible. Thus all moduli are at most one.

2. Their product is one and none exceeds one, so every modulus equals one.

3. Finiteness gives $z^{2^j} = z^{2^k}$ for some $j < k$. Since $z \neq 0$, $z^{2^j(2^{k-j}-1)} = 1$, proving the claim.

例题解析

1. 若某元素模大于一，反复平方在有限集中产生无界模，矛盾，因此所有模至多一。

2. 各模乘积为一且均至多一，故每个模均为一。

3. 有限性给出某个 $j < k$ 满足 $z^{2^j} = z^{2^k}$。由 $z \neq 0$，得 $z^{2^j(2^{k-j}-1)} = 1$，即为单位根。

The root map must come from the actual identity; closure under a different transformation gives no information.

根映射必须由原恒等式推出，不能擅自使用另一种封闭变换。

For the map $T(z) = 2z^3 + z$, a modulus $r > 1$ gives $|T(z)| \geq 2r^3 - r > r$, forcing unbounded growth. If a monic polynomial has constant term one, the product of root moduli is one. Hence all root moduli equal one; requiring the image modulus also one gives $|2z^2 + 1| = 1$, forcing $z^2 = -1$.

对 $T(z) = 2z^3 + z$，若模 $r > 1$，则 $|T(z)| \geq 2r^3 - r > r$，会迫使无界增长。首一且常数项为一的多项式，根模乘积为一，所以所有根模均为一；像的模也为一进一步给出 $|2z^2 + 1| = 1$，迫使 $z^2 = -1$。

Practice / 配套练习: AIME-2007-II-Q14 (WB p. 211)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 5 / 方法 5 [AL09-M05]

Use infinite integer descent to eliminate a nonzero difference

用无穷整数递降排除非零差

Procedure

1. Subtract adjacent cleared recurrences to isolate a sequence of integer differences.

2. Prove a nonzero difference forces the next difference nonzero but smaller in absolute value by a fixed factor.

3. An infinite sequence of positive integer magnitudes cannot decrease this way, so every difference must be zero.

4. Substitute the resulting periodicity back into the original recurrence and optimize only the remaining finite arithmetic choices.

通用步骤

1. 先清分母，再将相邻递推相减，分离整数差序列。

2. 证明非零差会迫使下一差也非零，但绝对值按固定比例缩小。

3. 正整数绝对值不能如此无穷递降，因此每个差都须为零。

4. 将所得周期代回原递推，再对剩余有限算术选择求最值。

> Example. An infinite integer sequence satisfies $b_n = (n+2)b_{n+1}$ for all $n \geq 0$. Prove it is identically zero.

> 例题：无穷整数序列满足 $b_n = (n+2)b_{n+1}$，$n \geq 0$，证明所有项均为零。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 5 / 方法 5 [AL09-M05] (continued / 续)

Worked solution

1. Iteration gives $b_0 = (k+1)!b_k$ for every $k \geq 1$.

2. If $b_0 \neq 0$, every b_k is a nonzero integer, so $|b_0| \geq (k+1)!$ for every k, impossible.

3. Thus $b_0 = 0$. The nonzero multipliers then force successively $b_1 = b_2 = \cdots = 0$.

例题解析

1. 反复代入得每个 $k \geq 1$ 都有 $b_0 = (k+1)!b_k$。

2. 若 $b_0 \neq 0$，则各 b_k 都是非零整数，迫使 $|b_0| \geq (k+1)!$ 对所有 k 成立，矛盾。

3. 因此 $b_0 = 0$，由各乘数非零，依次推出后续全零。

An inequality allowing the difference to become zero is insufficient; exact propagation of nonzeroness is essential.

若不等式允许差突然变零，就不能这样递降；必须证明非零性被精确传递。

Practice / 配套练习: AIME-2009-I-Q13 (WB p. 212)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 6 / 方法 6 [AL09-M06]

Normalize a radical recurrence into a folded angle map

将根式递推归一化为折叠角映射

Procedure

1. Remove an exponential scale so the radical becomes $\sqrt{1-u^2}$.

2. Choose a sine parameter $u=\sin\phi$ and write the coefficients as $\cos\theta,\sin\theta$.

3. Use $\sqrt{1-\sin^2\phi}=|\cos\phi|$, splitting at right-angle crossings.

4. Prove the eventual orbit or cycle, then restore the original scale.

通用步骤

1. 先除去指数尺度，使根式变为 $\sqrt{1-u^2}$。

2. 令 $u=\sin\phi$，将系数写成 $\cos\theta,\sin\theta$。

3. 必须使用 $\sqrt{1-\sin^2\phi}=|\cos\phi|$，在跨越直角时分类。

4. 证明最终轨道或循环，再恢复原尺度。

> Example. Let $u_0=0$ and $u_{n+1}=(\sqrt{3}/2)u_n+(1/2)\sqrt{1-u_n^2}$. Describe all terms from u_2 onward.

> 例题：令 $u_0=0,u_{n+1}=(\sqrt{3}/2)u_n+(1/2)\sqrt{1-u_n^2}$，描述从 u_2 起的全部项。

Worked solution

1. The first terms are $u_1=1/2,u_2=\sqrt{3}/2,u_3=1$, using the sine addition formula with angle 30°.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 6 / 方法 6 [AL09-M06] (continued / 续)

2. At $u = 1$, the next value is $\sqrt{3}/2$. At $u = \sqrt{3}/2$, the positive radical is $1/2$, giving the next value one.

3. Thus the exact two-cycle $\sqrt{3}/2, 1$ starts at u_2. Continuing to increase a formal angle without the absolute cosine would give the wrong later terms.

例题解析

1. 用三十度正弦和角公式，前几项为 $1/2, \sqrt{3}/2, 1$。

2. 当 $u = 1$ 时，下项为 $\sqrt{3}/2$；当 $u = \sqrt{3}/2$ 时，正根式为二分之一，下项为一。

3. 因此从 u_2 起恰交替为 $\sqrt{3}/2, 1$。若忽略余弦绝对值而持续增加形式角，后续会算错。

A nonnegative radical is an absolute value, so angle addition may turn into angle subtraction.

非负根式对应绝对值，因此和角过程可能改为差角。

Practice / 配套练习: AIME-2009-II-Q14 (WB p. 213)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07]

Count finite functions by intrinsic image and preimage levels

按内在像集与原像层次计数有限函数

Procedure

1. Translate a composition equality into restrictions on directed functional graphs.

2. Choose intrinsic classes such as fixed points, their direct preimages, and more distant points; the classes must be recoverable from the function.

3. Assign images only to their permitted classes, using independent powers after the classes are chosen.

4. If the image is prescribed exactly, require surjectivity rather than counting all maps into it.

通用步骤

1. 把复合等式转为有向函数图限制。

2. 选择能由函数自身恢复的类别，如不动点、其直接原像和更远点。

3. 选定类别后，只允许把像分配到指定层，再用独立幂计数。

4. 若像集须恰为某集合，必须要求满射，不能把所有映入函数都算入。

> Example. Count functions on a labeled three-element set satisfying $f^2 = f$.

> 例题：求固定标号三元集合上满足 $f^2 = f$ 的函数数。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07] (continued / 续)

Worked solution

1. Every image point must be fixed, and every fixed point belongs to the image. Thus choosing the image is exactly choosing the fixed-point set.

2. If it has $k \geq 1$ elements, choose it in $\binom{3}{k}$ ways. Each remaining element maps independently to any of its k points.

3. The count is $\sum_{k=1}^{3} \binom{3}{k} k^{3-k} = 3 + 6 + 1 = 10$. The fixed-point set is intrinsic, so no function is counted twice.

例题解析

1. 每个像点都必须不动，而每个不动点都属于像，因此像集恰为不动点集。

2. 若大小为 $k \geq 1$，有 $\binom{3}{k}$ 种选择，其余点各可独立映入其中任一点。

3. 总数为 $\sum_{k=1}^{3} \binom{3}{k} k^{3-k} = 10$。不动点集由函数唯一确定，故不重复。

A chosen target set is not automatically the exact image set.

指定允许像集并不等于指定准确像集。

If f^2 is constant a, let $S = f^{-1}(a)$. Then $a \in S$, every point of S maps to a, and each outside point maps to $S \setminus \{a\}$. For an n-point domain, a size-k set contributes $n\binom{n-1}{k-1}(k-1)^{n-k}$.

If $f^3 = f^2$, every second image is fixed. Choose i fixed points and j nonfixed direct preimages; all remaining points map to that middle layer. The count for those sizes is $\binom{n}{i}\binom{n-i}{j} i^j j^{n-i-j}$, with $0^0 = 1$ for an empty assignment.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07] (continued / 续)

The number of maps onto a specified k-set from an n-set is $\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n$. Multiply by the choice of the image set. A second map avoiding that image has $(n-k)^n$ choices.

若 f^2 为常值 a，令 $S=f^{-1}(a)$，则 $a\in S$，S 内各点映到 a，外点各映到 $S\setminus\{a\}$。在 n 元域中，大小 k 的贡献为 $n\binom{n-1}{k-1}(k-1)^{n-k}$。

若 $f^3=f^2$，每个第二像均不动。选 i 个不动点和 j 个非不动直接原像，其余点全映到中层；对应计数为 $\binom{n}{i}\binom{n-i}{j}i^j j^{n-i-j}$，空分配时约定 $0^0=1$。

从 n 元集满射到指定 k 元集的数目为 $\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n$，再乘像集选择数。第二个函数若避开该像集，则有 $(n-k)^n$ 种。

Practice / 配套练习: AIME-2013-II-Q11 (WB p. 215), AIME-2014-I-Q12 (WB p. 746), AIME-2018-II-Q10 (WB p. 207)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 8 / 方法 8 [AL09-M08]

Factor forced roots and apply finite-difference rigidity

分离被迫根并使用有限差分刚性

Procedure

1. Substitute special inputs into a polynomial identity to force explicit roots.

2. Factor those roots before canceling common polynomial factors in the identity.

3. A nonconstant degree-d polynomial has a nonzero first difference of degree $d-1$, so a periodic polynomial is constant.

4. Use normalization to determine the remaining constant and verify the original identity.

通用步骤

1. 向多项式恒等式代入特殊值，找出被迫根。

2. 先分离这些根，再在恒等式中约去公共多项式因子。

3. 非常数 d 次多项式的一阶差分为非零的 $d-1$ 次式，因此周期多项式只能为常数。

4. 由归一条件确定余下常数，并验证原恒等式。

> Example. Find all real polynomials P with $P(x+1)-P(x)=2x+1$.

> 例题：求满足 $P(x+1)-P(x)=2x+1$ 的所有实系数多项式。

Worked solution

1. The polynomial x^2 has the required first difference.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 8 / 方法 8 [AL09-M08] (continued / 续)

2. Put $Q(x) = P(x) - x^2$. Then $Q(x+1) - Q(x) = 0$, so Q is a periodic polynomial.

3. If Q were nonconstant, its first difference would have nonzero leading coefficient. Hence $Q = C$, and all solutions are $P = x^2 + C$.

例题解析

1. x^2 的一阶差分正好是 $2x + 1$。

2. 令 $Q(x) = P(x) - x^2$，则其差分为零，因此 Q 是周期多项式。

3. 若 Q 非常数，其差分首项系数非零，矛盾。因此 $Q = C$，全部解为 $P = x^2 + C$。

Cancel polynomial factors as identities; do not incorrectly exclude their zeros from the final polynomial.

约去因子是在多项式恒等式层面进行，不能把该因子的零点从最终多项式中排除。

Practice / 配套练习: AIME-2016-I-Q11 (WB p. 57)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 9 / 方法 9 [AL09-M09]

Periodic information and uniqueness force a fixed residue

周期信息与唯一性迫使固定余数

Procedure

1. Encode the full infinite itinerary by signs, interval labels, or digits.

2. Shift by one full itinerary period and prove the transformed point satisfies the same complete information.

3. Use uniqueness to equate the transformed and original point, giving a fixed-point congruence.

4. Test every finite candidate against the entire repeating itinerary and its domain.

通用步骤

1. 以符号、区间标记或数字编码完整无穷轨迹。

2. 移动一个信息周期，证明变换后的点仍满足同一套完整信息。

3. 利用唯一性令变换后点等于原点，得到不动余数关系。

4. 将每个有限候选代回整个周期轨迹及定义域检查。

> Example. For $F(x)=\{3x\}$ on $[0,1)$, find the point whose labels $\lfloor 3F^n(x)\rfloor$ alternate $0,2,0,2,\ldots$.

> 例题：对 $[0,1)$ 上的 $F(x)=\{3x\}$，求使标记 $\lfloor 3F^n(x)\rfloor$ 交替为零、二、零、二的点。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 9 / 方法 9 [AL09-M09] (continued / 续)

Worked solution

1. The labels give the ternary expansion $0.\overline{02}_3$. Its first $2k$ digits restrict x to an interval of length 9^{-k}, proving uniqueness.

2. After two shifts the same itinerary remains, so $F^2(x) = x$. Its two-digit integer part is two, hence $9x = 2 + x$ and $x = 1/4$.

3. The orbit is $1/4, 3/4, 1/4, \ldots$, whose labels are exactly zero and two, verifying existence.

例题解析

1. 标记给出三进制展开 $0.\overline{02}_3$，前 $2k$ 位将点限制在长 9^{-k} 的区间内，因此唯一。

2. 两次移位后信息不变，故 $F^2(x) = x$，两位整数部分为二，所以 $9x = 2 + x$，解得四分之一。

3. 轨道为四分之一、四分之三交替，标记恰为零、二，验证存在。

Uniqueness concerns the full infinite itinerary, not a short initial sign table.

唯一性针对完整无穷轨迹，不是有限开头的符号表。

Practice / 配套练习: AIME-2019-II-Q10 (WB p. 208)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10]

Propagate preimages branch by branch and prove their pattern

逐分支传播原像并证明其模式

Procedure

1. Determine the exact range of each branch before taking an inverse.

2. For an absolute value retain only nonnegative targets; include both signs and remove coinciding preimages.

3. Use induction to prove an arithmetic zero pattern or nested interval-index rule.

4. Count or sum only after the branch-domain restrictions and duplicates have been settled.

通用步骤

1. 求原像前，先确定每个分支的准确值域。

2. 绝对值只接受非负目标，取正负两支，并去除重合原像。

3. 以归纳证明等差零点模式或嵌套区间下标规则。

4. 只有明确分支定义域及重复后，才能计数或求和。

> Example. On $[0,1]$, let $g(x)=|2x-1|$. Find all zeros of $g\circ g\circ g$.

> 例题：在 $[0,1]$ 上令 $g(x)=|2x-1|$，求三次复合的全部零点。

Worked solution

1. The preimage of zero is the single point $1/2$; the two formal signs coincide there.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10] (continued / 续)

2. The preimages of $1/2$ are $1/4, 3/4$. For a target $t \in [0,1]$, its preimages are $(1-t)/2, (1+t)/2$.

3. Applying this to both targets gives $1/8, 3/8, 5/8, 7/8$. They are distinct and remain in the domain, so this is the complete zero set.

例题解析

1. 零的原像只有二分之一，此时形式上的正负分支重合。

2. 二分之一的原像为四分之一、四分之三；一般 $t \in [0,1]$ 的原像为 $(1-t)/2, (1+t)/2$。

3. 对两目标再求原像，得到八分之一、八分之三、八分之五、八分之七，互异且均在域内，故已穷尽。

A negative intermediate zero has no preimage under a nonnegative absolute-value map.

在非负绝对值映射下，负的中间零点没有原像。

For $f(x) = x\{x\}$ on $[j, j+1)$ with $j \geq 0$, write $x = j + t$. The expression $(j+t)t$ is strictly increasing with range $[0, j+1)$. Thus a positive target has exactly one preimage in each admissible unit interval. Repeated inverses produce weakly ordered interval indices, counted by binomial coefficients.

For $F_1(x) = |x-1|, F_n(x) = F_{n-1}(|x-n|)$, the zero set is the step-two progression from $n - T_{n-1}$ to T_n, where $T_n = n(n+1)/2$. Inductively the nonnegative preceding zeros include every value of their parity from zero or one to T_{n-1}; reflecting these as $n \pm z$ produces exactly the stated progression. Its endpoints sum to $2n$ and it has $T_{n-1} + 1$ points.

对 $j \geq 0$ 的区间 $[j, j+1)$，写 $x = j + t$，则 $x\{x\} = (j+t)t$ 严格递增，值域为 $[0, j+1)$。因此每个允许单位区间内，正目标恰有一个原像；反复求逆得到弱递增区间下标，可用组合数计数。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10] (continued / 续)

若 $F_1(x)=|x-1|, F_n(x)=F_{n-1}(|x-n|)$，其零点为从 $n-T_{n-1}$ 到 T_n 的公差二数列，其中 $T_n=n(n+1)/2$。归纳时，前一层非负零点含从零或一到 T_{n-1} 的全部同奇偶值；反射为 $n\pm z$ 后恰填满该数列。两端和为 $2n$，点数为 $T_{n-1}+1$。

Practice / 配套练习: AIME-2020-II-Q08 (WB p. 209), AIME-2020-II-Q14 (WB p. 178)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 11 / 方法 11 [AL09-M11]

Unroll a recursive function to its first reachable terminal input

展开递归直到首个可达终止输入

Procedure

1. Identify the terminal set and the arithmetic progression of inputs visited by the recursion.

2. Prove a terminal point is reachable and choose the first one in that progression, not merely the nearest terminal numerically.

3. Add the cost of each recursive step to the terminal value.

4. For a minimum input, combine the terminal formula with its defining interval and residue constraints.

通用步骤

1. 确定终止集合，以及递归访问的输入等差数列。

2. 证明终止点可达，选择该等差数列中的首个终止点，而非数值最近的任意终止点。

3. 将每次递归的代价加上终止值。

4. 若求最小输入，将终止公式与其定义区间、余数限制合并。

> Example. For positive integers define $H(n) = 0$ if $5 \mid n$, and $H(n) = 1 + H(n+2)$ otherwise. Find $H(7)$.

> 例题：正整数上定义 $H(n)$：若五整除 n 则为零，否则为 $1 + H(n+2)$，求 $H(7)$。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 11 / 方法 11 [AL09-M11] (continued / 续)

Worked solution

1. The visited inputs are $7+2j$. Because two is invertible modulo five, a multiple of five is reachable.

2. The first nonnegative solution of $7+2j \equiv 0 \pmod 5$ is $j=4$, giving input fifteen.

3. The four preceding calls each add one, so $H(7)=4$. The closer multiple ten is not visited and must not be used.

例题解析

1. 访问输入为 $7+2j$，因二在模五下可逆，必能到达五的倍数。

2. 同余式的首个非负解为 $j=4$，终止输入十五。

3. 此前四次调用各加一，所以结果为四；较近的十不在访问序列中，不能用它。

A square reached by step two must have the input parity; the first square above the input may be skipped.

步长二只能到达与输入同奇偶的平方，紧邻上方的平方可能被跳过。

Let $k=\lceil\sqrt{n}\rceil$. Unit-step recursion first reaches k^2. A step-two recursion first reaches k^2 if $n \equiv k \pmod 2$, otherwise $(k+1)^2$. With cost two per step, its accumulated cost equals terminal square minus n. Always restore $(k-1)^2 < n \le k^2$ before minimizing.

令 $k=\lceil\sqrt{n}\rceil$，步长一首先到 k^2；步长二若 $n \equiv k \pmod 2$ 也到 k^2，否则首先到 $(k+1)^2$。若每步代价为二，累计代价恰为终止平方减去 n。求最小值时必须恢复 $(k-1)^2 < n \le k^2$。

Practice / 配套练习: AIME-2021-II-Q15 (WB p. 216)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 12 / 方法 12 [AL09-M12]

Conjugate a rational map to a simpler recurrence

将有理映射共轭为较简单递推

Procedure

1. Search for a fractional-linear variable that sends special points or fixed points to simple values.

2. Derive the transformed recurrence exactly and invert the substitution.

3. Track numerator and denominator before and after reduction; use gcd identities to certify the claimed denominator growth.

4. Prove the transformed orbit remains in the interval that gives valid positive original values before taking modular residues.

通用步骤

1. 寻找将特殊点或不动点送到简单数值的分式线性变量。

2. 精确推出变换后的递推，并写出逆变换。

3. 跟踪约分前后的分子分母，用最大公因数关系确认所称分母增长。

4. 取同余前，证明变换轨道留在使原变量为合法正值的区间内。

> Example. Let $x_0 = 3$ and $x_{n+1} = (x_n^2 + 1)/(2x_n)$. Find an exact reduced formula for x_n.

> 例题：令 $x_0 = 3, x_{n+1} = (x_n^2 + 1)/(2x_n)$，求 x_n 的精确既约表达式。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 12 / 方法 12 [AL09-M12] (continued / 续)

Worked solution

1. Put $y = (x-1)/(x+1)$. Direct simplification gives $y_{n+1} = y_n^2$ and $y_0 = 1/2$.

2. Thus $y_n = 2^{-2^n}$, always in $(0,1)$, and inversion gives $x_n = (2^{2^n}+1)/(2^{2^n}-1)$.

3. The two numerator/denominator terms differ by two and are both odd. Their gcd is therefore one, so the formula is reduced and positive.

例题解析

1. 令 $y = (x-1)/(x+1)$，直接化简得 $y_{n+1} = y_n^2, y_0 = 1/2$。

2. 因此 $y_n = 2^{-2^n}$ 始终在零一之间，逆变换给出 $x_n = (2^{2^n}+1)/(2^{2^n}-1)$。

3. 分子分母相差二且均奇，最大公因数只能为一，所以该式已既约且为正。

A denominator recurrence is not proved until cancellation has been ruled out.

只有排除约分后，分母递推才算证明。

For a transformed map $y \mapsto y(y-1)$ with $y = c/d$ in lowest terms, the unreduced next fraction is $c(c-d)/d^2$. If $\gcd(c,d) = 1$, both factors in the numerator are coprime to d, so the next denominator is exactly d^2. When inverting a substitution, analyze any new common divisor separately.

若变换映射为 $y \mapsto y(y-1)$，既约 $y = c/d$ 的下一式为 $c(c-d)/d^2$。由 $\gcd(c,d) = 1$，分子两因子都与 d 互质，故下一分母准确为 d^2。逆变换若引入新的公因子，还须另行核查。

Practice / 配套练习: AIME-2025-II-Q13 (WB p. 217)